

NITHYA MADALAKOTE PRAKASH

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PROFESSIONAL SUMMARY

AI Engineer pursuing a Master's in Artificial Intelligence in Germany, focused on turning research and modern AI techniques into practical, production-ready software rather than notebook-only demos. Experienced across LLM applications, Retrieval-Augmented Generation, AI agents, NLP, computer vision, machine learning, backend engineering, and MLOps/deployment, with particular attention to reliability, evaluation, scalability, observability, and clean architecture. Currently building production-ready AI systems, RAG applications, and intelligent backend products independently.

EDUCATION

M.Sc. Artificial Intelligence — Brandenburgische Technische Universität (BTU) Cottbus

Cottbus, Germany | 2025 – Present

- Focus areas: Artificial Intelligence, Machine Learning, Generative AI, NLP, Computer Vision

B.Tech, Computer Science — Presidency University

Bengaluru, India | 2020 – 2024

- Strong foundation in algorithms, data structures, software engineering, and applied machine learning

WORK EXPERIENCE

Java Full Stack Developer Trainee — Palle Technologies

Bengaluru, India | Jun 2024 – Jan 2025

- Completed a structured 6-month full stack development programme, gaining experience in Java, Spring Boot, MySQL, and responsive front-end development.
- Built and tested REST APIs and integrated front-end and back-end components for internal web applications.
- Demonstrated strong problem-solving, code review participation, and team collaboration across the SDLC.

TECHNICAL SKILLS

AI / ML: Python, PyTorch, scikit-learn, Transformers, NLP, Computer Vision, Machine Learning, Deep Learning

Generative AI: LLMs, RAG, Embeddings, Vector Search, Hybrid Retrieval, Reranking, AI Agents, Prompt Engineering, LLM Evaluation

Backend: Python, FastAPI, REST APIs, PostgreSQL, Redis, Celery

Infrastructure: Docker, Git, GitHub, Linux, CI/CD, MLOps

Frontend: Next.js, React, TypeScript, Tailwind CSS

Databases / Data: PostgreSQL, Qdrant, SQLite, SQL

PROJECTS

Production RAG Knowledge Assistant | github.com/nithya-prakash/HybridRAG

Stack: Python, FastAPI, Next.js, Qdrant, PostgreSQL, Redis, Celery, Docker, LLMs, Embeddings

- Built a production-oriented enterprise knowledge assistant using Retrieval-Augmented Generation to answer questions over private documents with grounded, cited responses.
- Implemented hybrid retrieval combining semantic search and BM25, with reranking and citation-aware generation.
- Added background document processing with an evaluation harness and local LLM support, in a production-grade architecture end to end.

Predictive Maintenance & Anomaly Detection Platform |

github.com/nithya-prakash/PredictiveMaintenanceAI

Stack: Python, FastAPI, PyTorch, scikit-learn, PostgreSQL, Docker, Time-series ML

- Built an industrial AI platform for detecting abnormal machine behavior and supporting predictive maintenance workflows.
- Implemented sensor data processing, predictive modeling, and anomaly detection with machine health monitoring.
- Added alerting and automated data pipelines for real telemetry.

Industrial Multimodal AI Copilot | github.com/nithya-prakash/IndustrialAICopilot

Stack: Python, FastAPI, Celery, PostgreSQL, Qdrant, Docker, Prometheus

- Built an evidence-based diagnostic copilot for manufacturing technicians, combining a technician's question, a component photo, sensor readings, and technical manuals into a structured, cited diagnosis with confidence scoring.
- Fused RAG over manuals, computer vision, and live sensor analytics in a multimodal pipeline with agentic orchestration and human-in-the-loop approval for high-risk cases.
- Built and verified phase-by-phase against real Postgres/Qdrant/Docker infrastructure.

SipMate / Hydra | github.com/nithya-prakash/SipmateAssistant

Stack: Python, PySide6, PyObjC, SQLite, scikit-learn, macOS

- Built a macOS desktop hydration assistant combining a playful interface with local data storage and ML-based scheduling logic.
- Implemented native macOS integration with procedural characters and a custom desktop UI in PySide6.

DeutschAI | github.com/nithya-prakash/DeutschAI

Stack: Python, FastAPI, Next.js, LLM Tutor Agent

- Built an AI-powered, adaptive platform for learning German from A1 to C1, architected for additional languages to be added without reworking the core.
- Implemented a spaced-repetition learning engine paired with a retrieval-grounded AI tutor, local speech practice, and ML-driven analytics and recommendations.

GreenGrid Optimizer | github.com/nithya-prakash/GreenGridOptimizer

Stack: Python, Machine Learning, Time Series Forecasting

- Built an end-to-end ML application forecasting renewable energy generation across Germany using public datasets.

PUBLICATIONS

MediSync – Streamlining Healthcare Continuity

International Journal of Research Publication and Reviews (IJRPR), 2023 | ISSN 2582-7421

- A doctor appointment web app (JavaScript, HTML, CSS, Bootstrap, MySQL, Spring Boot) connecting patients and doctors directly — search for a doctor, book an appointment, and receive reminder notifications, with no third party involved.